

OmniXRI TinyML 小學堂 2025



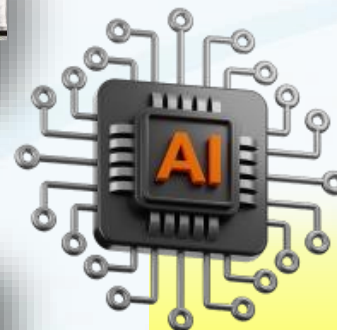
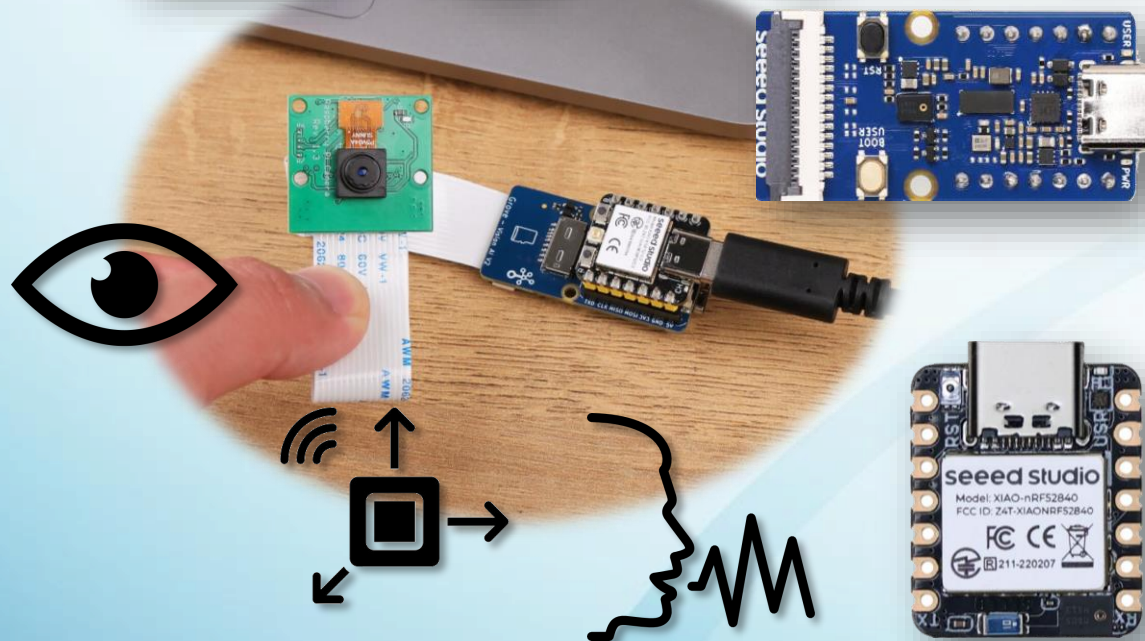
歡迎加入
邊緣人俱樂部



沒有最邊



只有更邊



Cortex-M
Processor

Ethos-U
MicroNPU

【第 12 講】

影像應用—姿態估測



歐尼克斯實境互動工作室 (OmniXRI Studio)
許哲豪 (Jack Hsu)

簡報大綱



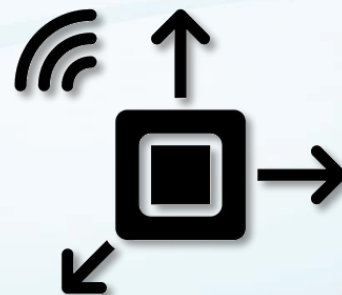
- 12.1. Himax SDK 開發環境建置
- 12.2. Yolov8n 姿態資料集與模型
- 12.3. 程式編譯
- 12.4. 模型部署與測試

本課程完全免費，請勿移作商業用途！

歡迎留言、訂閱、點讚、轉發，讓更多需要的朋友也能一起學習。

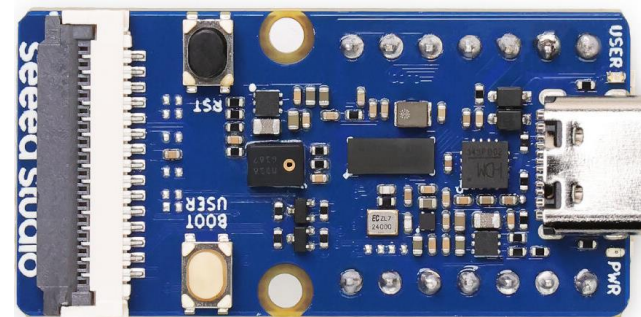
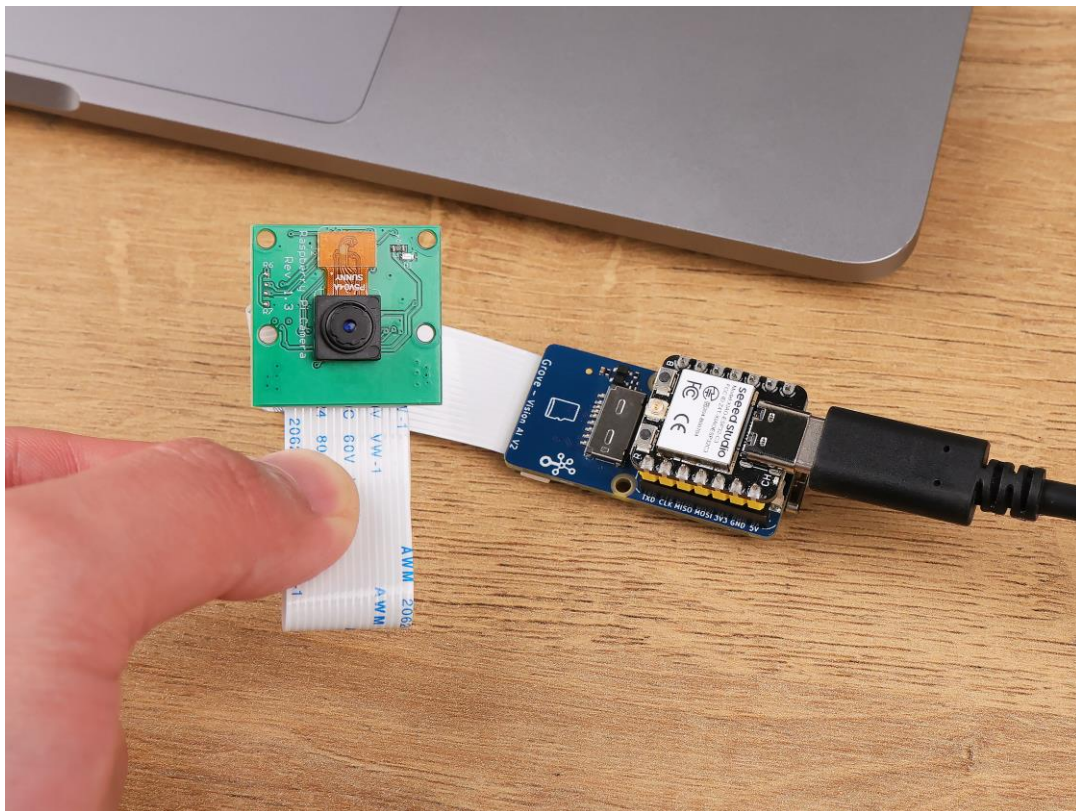
完整課程大綱：<https://omnixri.blogspot.com/2025/03/omnixri-tinymml-2025-0.html>

課程直播清單：<https://www.youtube.com/@omnixri1784/streams>

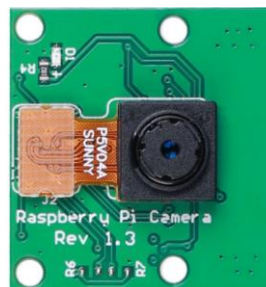


12.1. Himax SDK 開發環境建置

Seeed Grove Vision AI V2 Kit



**Grove Vision AI
Module V2**
(Himax WiseEye2
HX6538)



**OV5647-62
Camera Module
for Pi3B+4B**
(含軟排)

2529 x 1944 (5MP)

**Xiao ESP32C3
(WIFI / BT5)**
(無線通訊用，選配)



資料來源：https://wiki.seeedstudio.com/grove_vision_ai_v2/

Grove Vision AI Module V2 技術文件



https://wiki.seesstudio.com/grove_vision_ai_v2/

TinyML 開發流程選項

模型訓練工具名稱

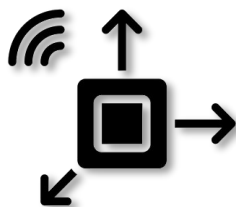
適用感測器

主要特色



TensorFlow Lite

參考【第 8 講】聲音辨識應用—環境音辨識

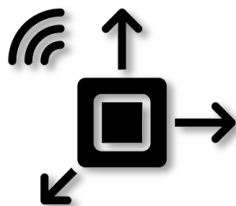


免費版：資料集需自行建置，僅提供模型訓練及轉成 TensorFlow Lite for Microcontroller 格式。可利用雲端 Google Colab 或本地端 Jupyter Notebooks 進行模型訓練及部署。



EDGE
IMPULSE

參考【第 6 講】TinyML 開發流程、
【第 11 講】影像應用—物件偵測



免費版：可支援各式感測器及開發板，提供一定額度資料集建置、資料前處理（特徵提取）模型訓練、模型優化及多種部署方式。亦可導入已訓練好的 TensorFlow Lite, ONNX 等模型。

SenseCraft



參考【第 10 講】影像應用—影像分類

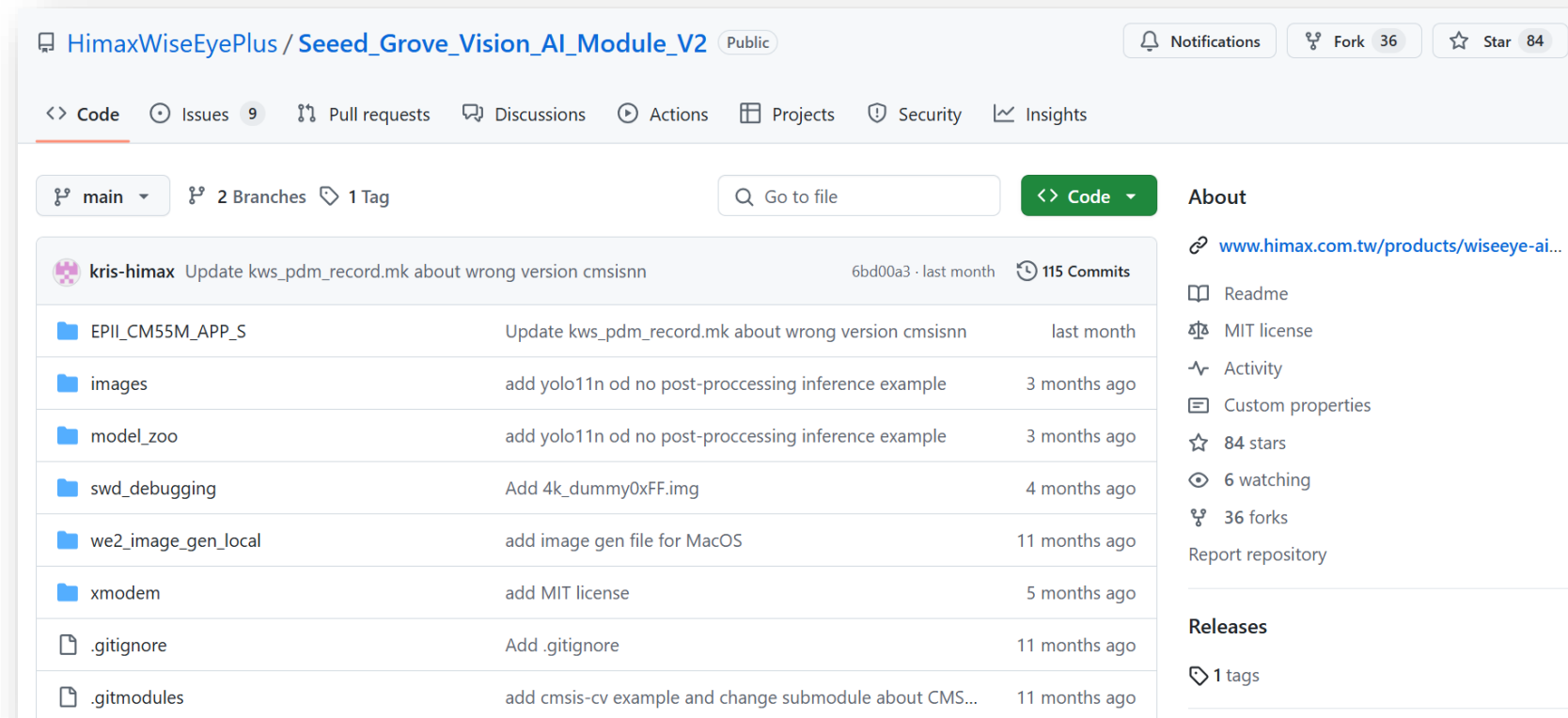
免費版：影像分類可多分類、物件偵測同一模型僅能一類。事件觸發只能設一種條件。可支援雲端資料收集、模型訓練及部署，亦導入外部已訓練好之 TensorFlow Lite 模型。

Github - Himax - Grove Vision AI Module V2

建置 elf

轉換 img

燒錄 MCU



https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2

Yolov8n Pose 參考文件及預編譯模型

scenario_app	project name
face mesh	tflm_fd_fm
yolov8n object detection	tflm_yolov8_od
yolov8n pose	tflm_yolov8_pose
yolov8n gender classification	tflm_yolov8_gender_cls
pdm mic record	pdm_record
KeyWord Spotting using Transformers	kws_pdm_record
imu read	imu_read
peoplenet from TAO	tflm_peoplenet
yolo11n object detection	tflm_yolo11_od

Seed_Grove_Vision_AI_Module_V2 / model_zoo /

模型動物園

 kris-himax add yolo11n od no post-processing info

Name

..

*.tflite

kws_pdm_record

tflm_fd_fm

tflm_yolo11_od

tflm_yolov8_gender_cls

tflm_yolov8_od

tflm_yolov8_pose

資料來源：

https://github.com/HimaxWiseEyePlus/Seed_Grove_Vision_AI_Module_V2/blob/main/EP11_CM55M_APP_S/app/scenario_app/tflm_yolov8_pose/README.md

準備 Windows 編譯環境 (1/2)

1. 安裝 **make** (參考 <https://github.com/xpack-dev-tools/windows-build-tools-xpack/releases>) 下列方式二選一
 - 先安裝 **xpm** 再執行 `xpm install @xpack-dev-tools/windows-build-tools@4.4.1-3.1 --verbose`
 - 直接下載 **xPack Windows Build Tools v4.4.1-3.zip** ，解壓縮到自定義路徑，假設在 **x:\xpack** 下，再將 **x:\xpack\xPack\xpack-windows-build-tools-4.4.1-3-win32-x64\xpack-windows-build-tools-4.4.1-3\bin** 手動加到環境變數下即可。
2. 下載 **Arm GNU Toolchain** [arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi.zip](https://github.com/ARM-software/infocenter/blob/master/html/Infocenter_V10.2/Toolchains/Arm_GNU_Toolchain/Arm_GNU_Toolchain_13.2_rel1_mingw-w64_i686_arm-none-eabi.zip) (347MB)
3. 解壓縮到自定義路徑，假設在 **x:\arm** 下

資料來源：https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2

準備 Windows 編譯環境 (2/2)

4. 增加工作路徑()到環境變數中，可手動增加或執行下列指令增加
`setx PATH "%PATH%;x:\arm\arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi\bin"`
5. 切換到自定義路徑，假設 `x:\himax\`，執行下列指令，複製下載 github repository

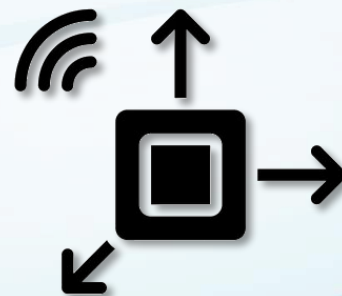
```
cd x:\himax
```

```
git clone --recursive
```

```
https://github.com/HimaxWiseEyePlus/Seeed\_Grove\_Vision\_AI\_Module\_V2.git
```

注意如果有多個版本的 **arm-gnu-toolchain**，要將環境變數的順序設前面一些，以免影響到編譯結果。

資料來源：https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2

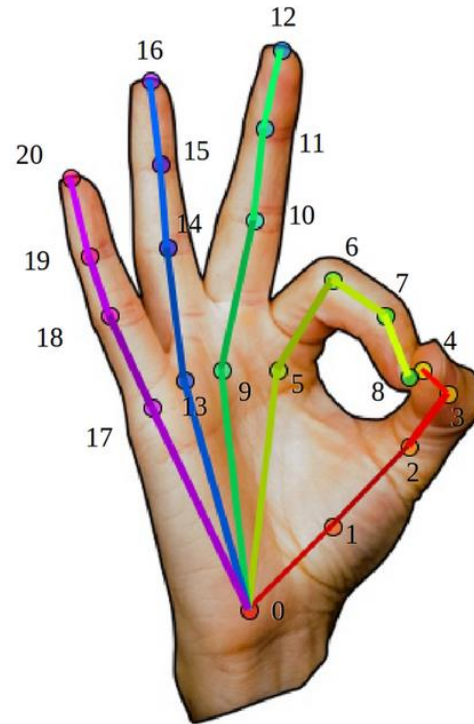


12.2. YOLOv8n 姿態資料集與模型

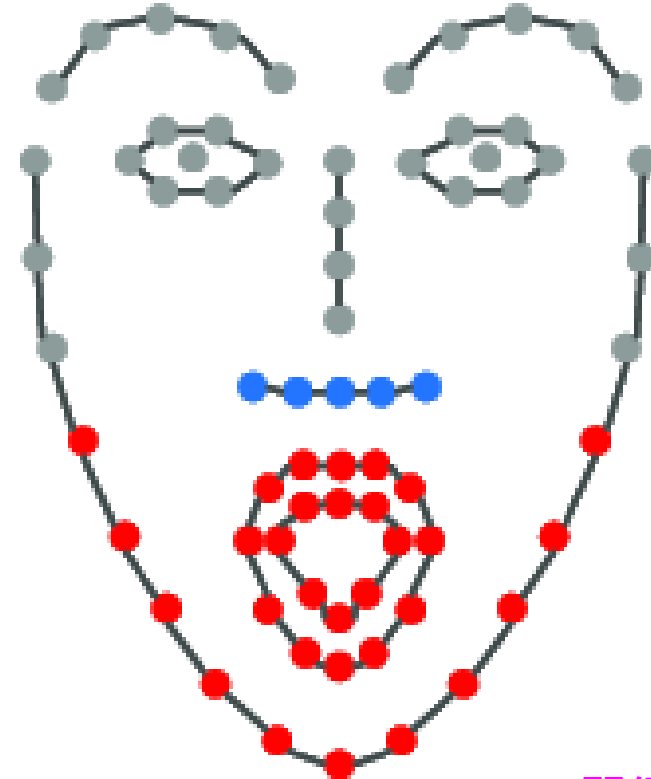
常見姿態估測資料集 – OpenPose



(a) body 18 關鍵點
(25 關鍵點)



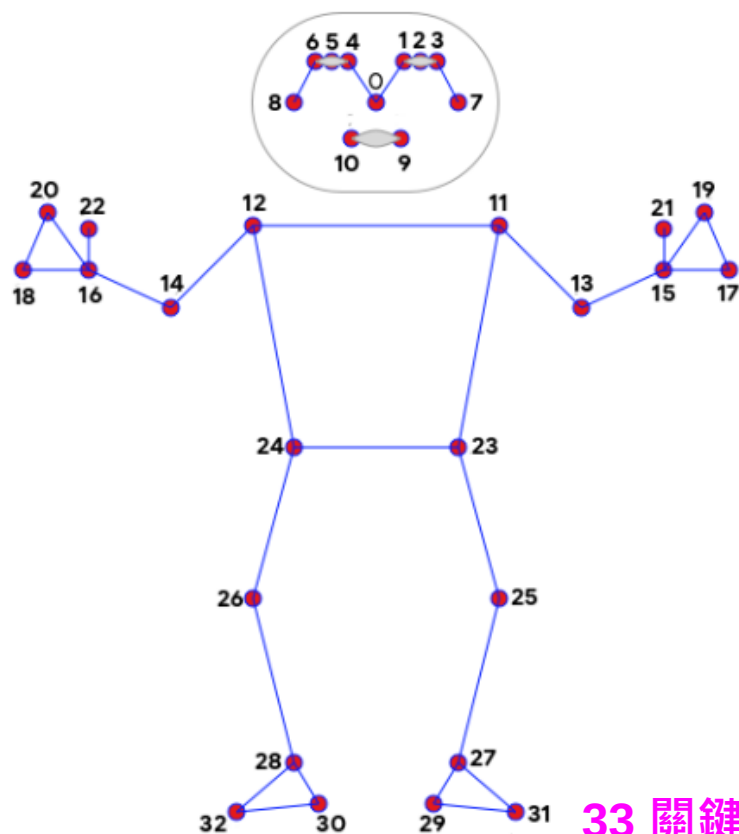
(b) hand 21 關鍵點



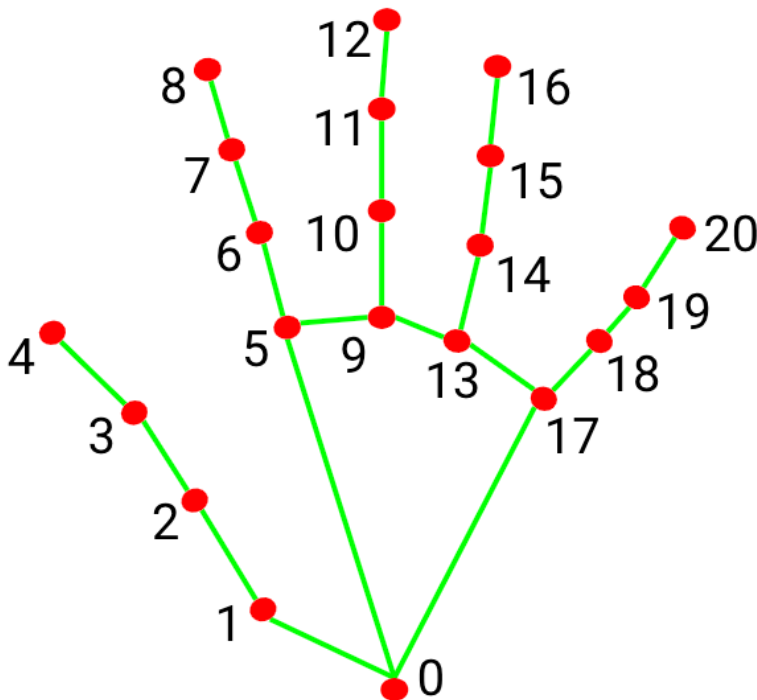
68 關鍵點

<https://github.com/CMU-Perceptual-Computing-Lab/openpose>

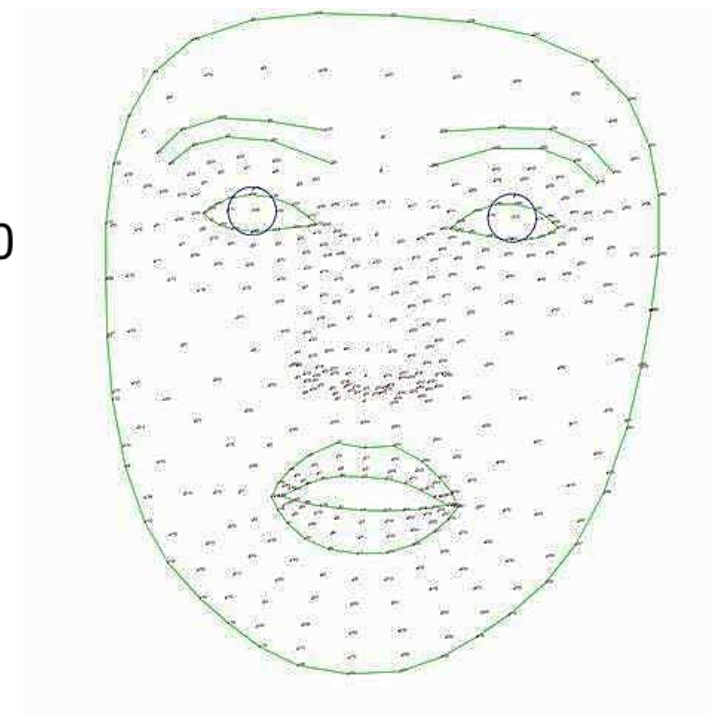
常見姿態估測資料集 – Google MediaPipe



33 關鍵點



21 關鍵點

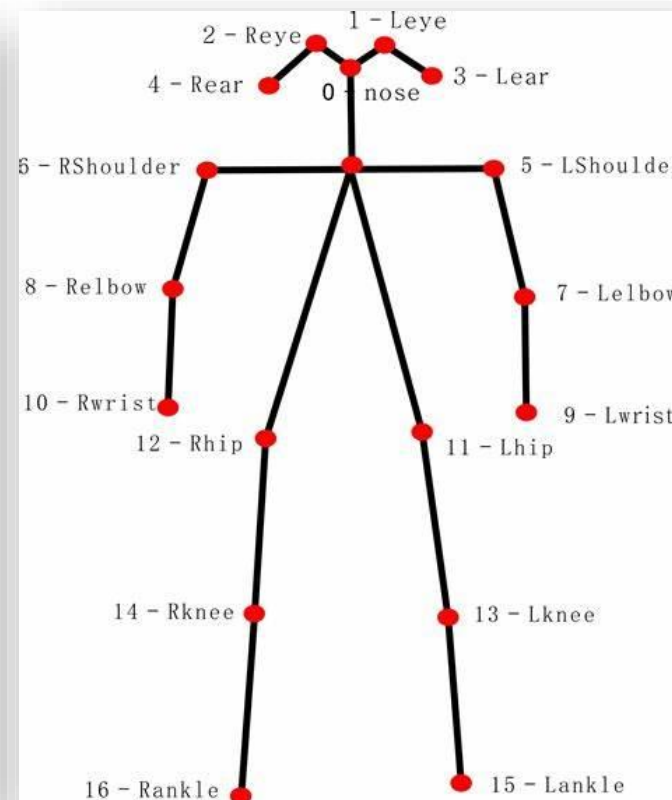


468 關鍵點
(3D 網格)

<https://ai.google.dev/edge/mediapipe/solutions/guide?hl=zh-tw>

常見姿態估測資料集 – Microsoft COCO

COCO (Common Objects in Context) 提供人體姿態關鍵點，超過20萬張已標註影像，25萬個關鍵點，每個人物有17個關鍵點。



- 0 鼻子
- 1 左眼
- 2 右眼
- 3 左耳
- 4 右耳
- 5 左肩
- 6 右肩
- 7 左肘
- 8 右肘
- 9 左腕
- 10 右腕
- 11 左髖
- 12 右髖
- 13 左膝
- 14 右膝
- 15 左踝
- 16 右踝

<https://cocodataset.org/#keypoints-2017>

Yolov8 主要功能

Ultralytics 提出之算法，另外 Yolov5, Yolov11 皆是同公司產品。



Classify
影像分類



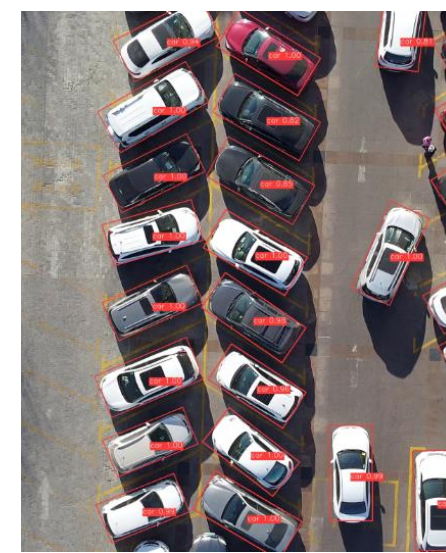
Detect
物件偵測



Segment
影像分割



Pose
姿態估測



Obb
旋轉物件偵測

資料來源：https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2

Yolov8 模型名稱

	Model	Filenames	Task	Inference	Validation	Training	Export
物件偵測	YOLOv8	yolov8n.pt yolov8s.pt yolov8m.pt yolov8l.pt yolov8x.pt	Detection	✓	✓	✓	✓
影像分割	YOLOv8-seg	yolov8n-seg.pt yolov8s-seg.pt yolov8m-seg.pt yolov8l-seg.pt yolov8x-seg.pt	Instance Segmentation	✓	✓	✓	✓
姿態估測	YOLOv8-pose	yolov8n-pose.pt yolov8s-pose.pt yolov8m-pose.pt yolov8l-pose.pt yolov8x-pose.pt yolov8x-pose-p6.pt	Pose/Keypoints	✓	✓	✓	✓
旋轉物件偵測	YOLOv8-obb	yolov8n-obb.pt yolov8s-obb.pt yolov8m-obb.pt yolov8l-obb.pt yolov8x-obb.pt	Oriented Detection	✓	✓	✓	✓
影像分類	YOLOv8-cls	yolov8n-cls.pt yolov8s-cls.pt yolov8m-cls.pt yolov8l-cls.pt yolov8x-cls.pt	Classification	✓	✓	✓	✓

資料來源：<https://docs.ultralytics.com/zh/models/yolov8/>

YOLOv8 姿態估測模型及效能

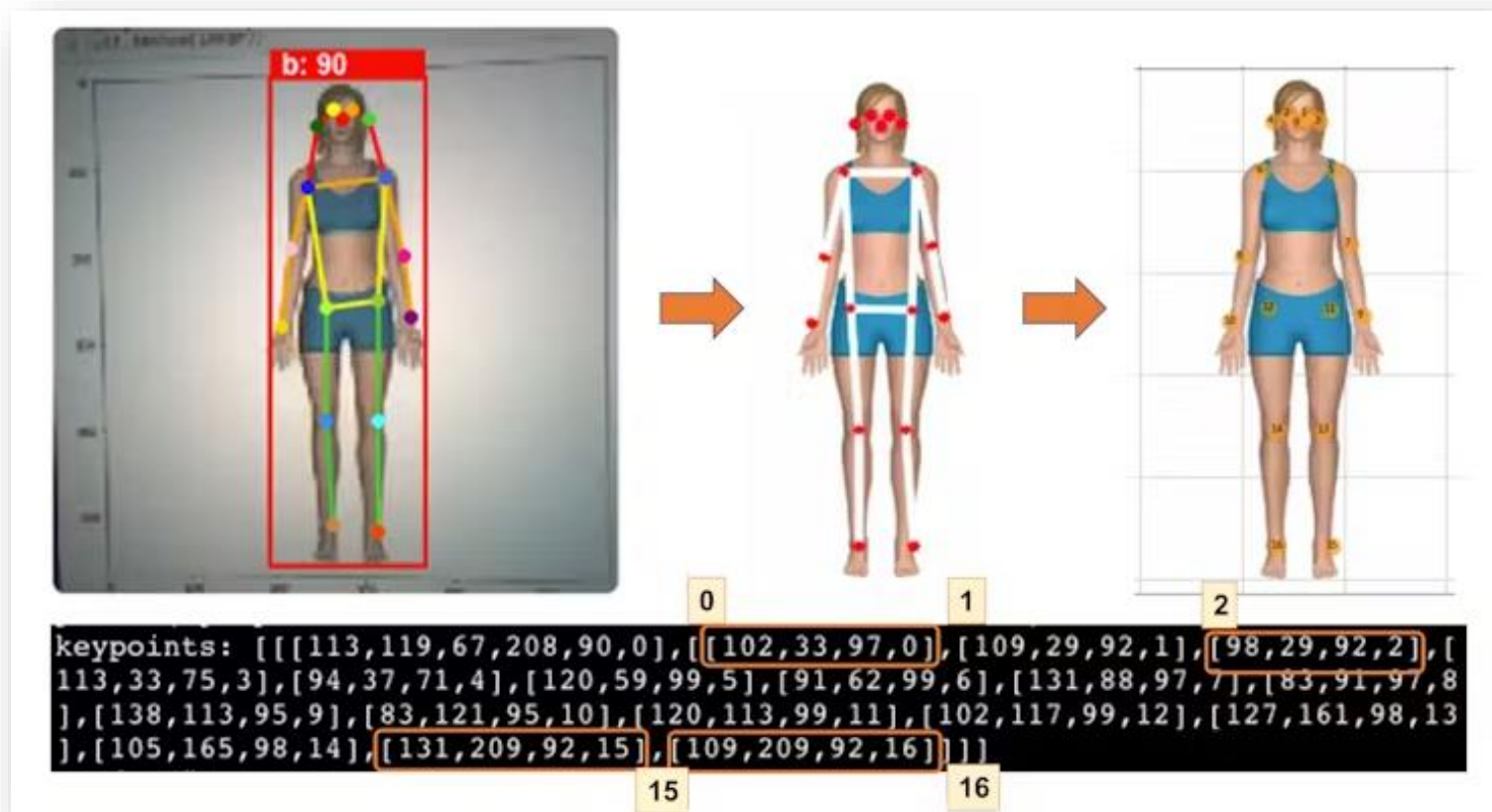
姿態輸出採 **COCO** 17 關鍵點定義。

模型	尺寸 (像素)	50-95	mAPpose 50	速度 CPU ONNX (毫秒)	速度 A100 TensorRT (毫秒)	params (M)	FLOPs (B)
YOLOv8n-姿勢	640	50.4	80.1	131.8	1.18	3.3	9.2
YOLOv8s-姿勢	640	60.0	86.2	233.2	1.42	11.6	30.2
YOLOv8m-姿勢	640	65.0	88.8	456.3	2.00	26.4	81.0
YOLOv8l-姿勢	640	67.6	90.0	784.5	2.59	44.4	168.6
YOLOv8x-姿勢	640	69.2	90.2	1607.1	3.73	69.4	263.2
YOLOv8x-pose- p6	1280	71.6	91.2	4088.7	10.04	99.1	1066.4

<https://docs.ultralytics.com/zh/models/yolov8/#performance-metrics>

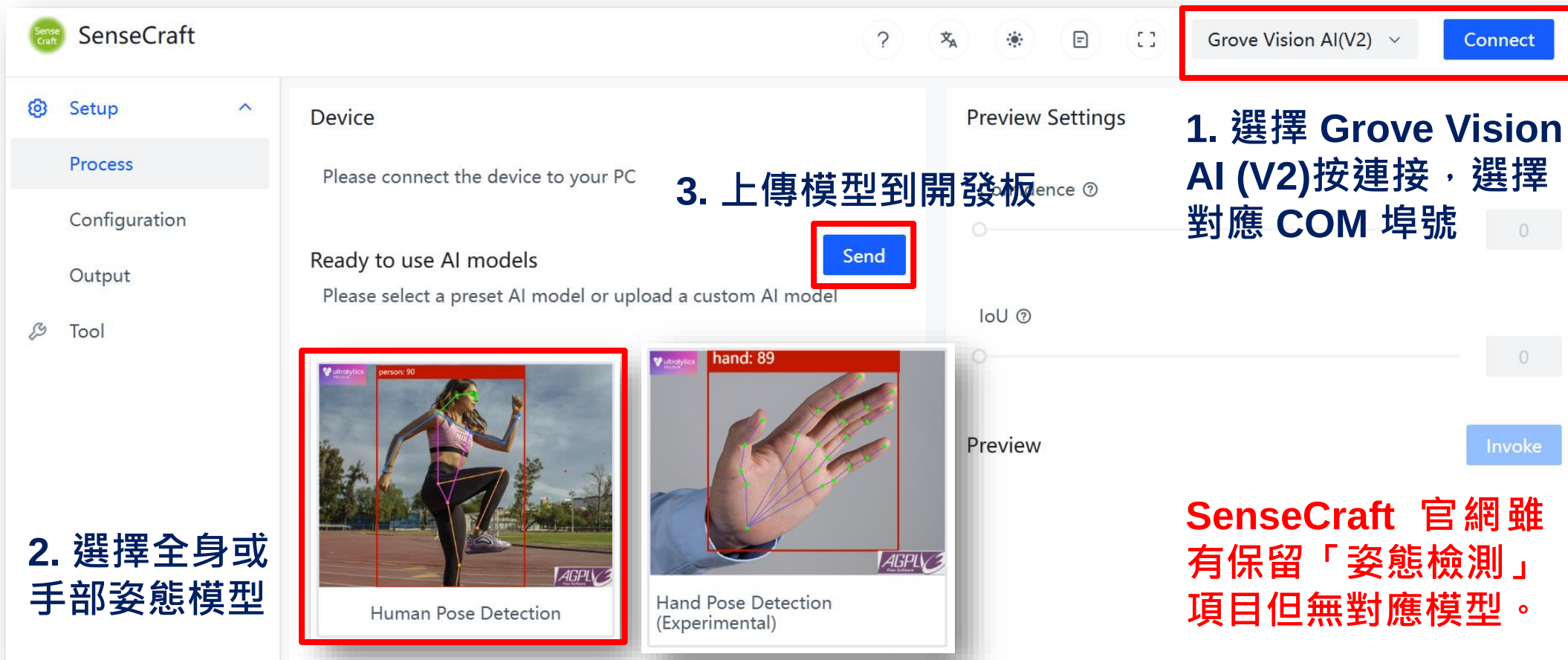
YOLOv8 姿態輸出結果

輸出結果：一個物件框，包含物件框中心座標(X,Y)、尺寸(寬, 高)、置信度、標籤，17個關鍵點，包含座標(X,Y)、置信度及關節編號。



資料來源：<https://www.hackster.io/mjrobot/computer-vision-at-the-edge-with-grove-vision-ai-module-v2-0003c7>

SenseCraft Web Toolkit – 姿態估測



The screenshot shows the SenseCraft Web Toolkit interface. On the left is a sidebar with navigation links: Setup, Process (highlighted), Configuration, Output, and Tool. The main area is divided into sections: 'Device' with a 'Send' button, 'Ready to use AI models' with two model cards ('Human Pose Detection' and 'Hand Pose Detection (Experimental)'), and 'Preview Settings' with sliders for 'Confidence' and 'IoU', and an 'Invoke' button. A red box highlights the 'Grove Vision AI(V2)' dropdown and 'Connect' button in the top right. Another red box highlights the 'Send' button. A third red box highlights the 'Human Pose Detection' model card. Blue and red text annotations provide instructions and notes.

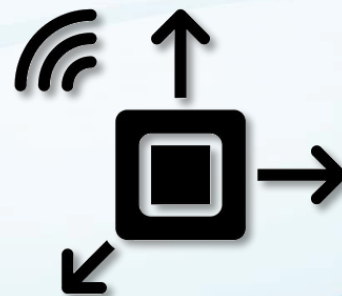
2. 選擇全身或手部姿態模型

3. 上傳模型到開發板

1. 選擇 Grove Vision AI (V2)按連接，選擇對應 COM 埠號

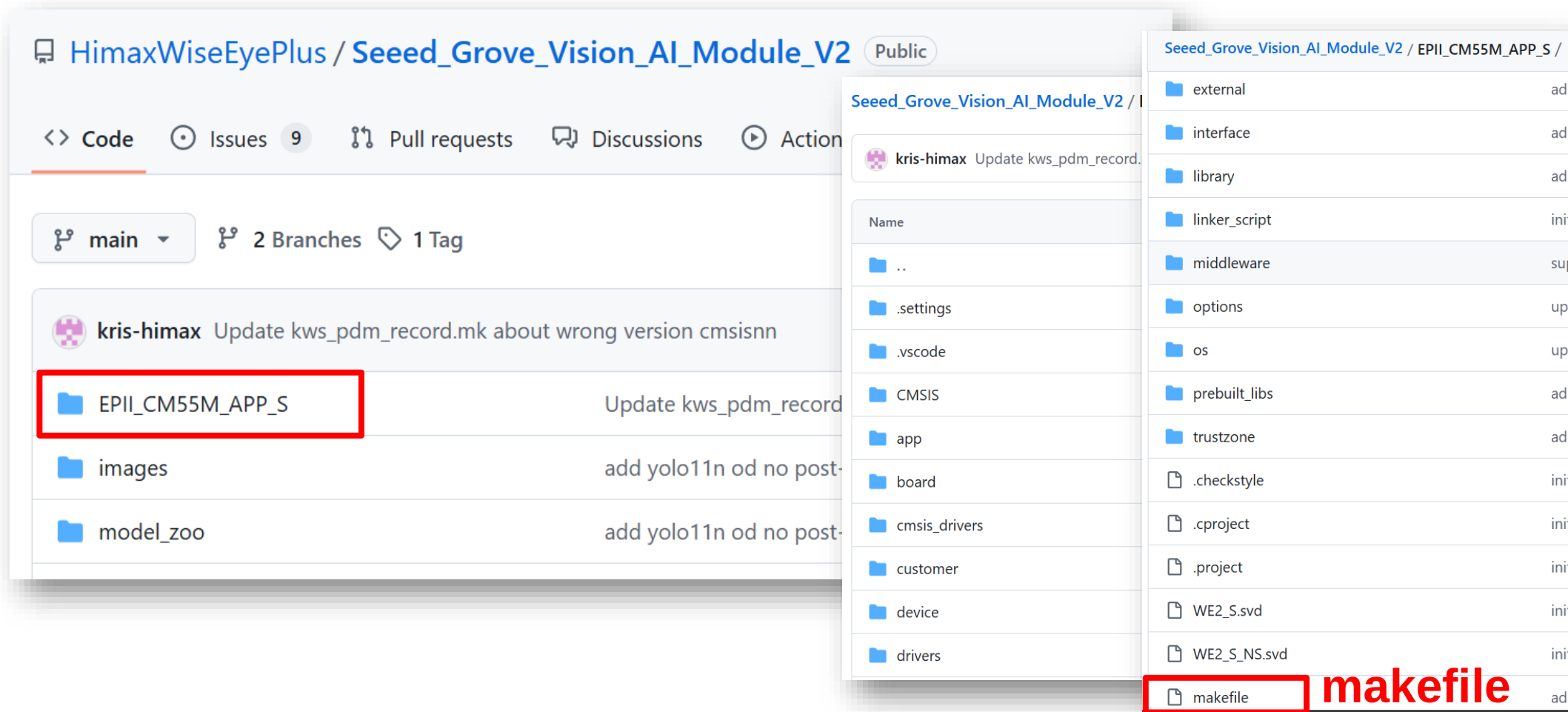
SenseCraft 官網雖有保留「姿態檢測」項目但無對應模型。

<https://seeed-studio.github.io/SenseCraft-Web-Toolkit/>



12.3. 程式編譯

EPII_CM55M_APP_S 待編譯程式



HimaxWiseEyePlus / Seeed_Grove_Vision_AI_Module_V2 Public

<> Code Issues 9 Pull requests Discussions Action

main 2 Branches 1 Tag

kris-himax Update kws_pdm_record.mk about wrong version cmsisnn

EPII_CM55M_APP_S Update kws_pdm_record

images add yolo11n od no post-

model_zoo add yolo11n od no post-

Seeed_Grove_Vision_AI_Module_V2 /

kris-himax Update kws_pdm_record

Name

- ..
- .settings
- .vscode
- CMSIS
- app
- board
- cmsis_drivers
- customer
- device
- drivers

Seeed_Grove_Vision_AI_Module_V2 / EPII_CM55M_APP_S /

- external
- interface
- library
- linker_script
- middleware
- options
- os
- prebuilt_libs
- trustzone
- .checkstyle
- .cproject
- .project
- WE2_S.svd
- WE2_S_NS.svd
- makefile

makefile

https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2

如何在 Windows 環境編譯 (1/2)

1. 進入工作路徑

```
cd Seeed_Grove_Vision_AI_Module_V2
```

2. 修改 makefile 內容，變更 APP_TYPE 為 **tflm_yolov8_pose**

```
APP_TYPE = tflm_yolov8_pose
```

3. 編譯程式

```
cd EPII_CM55M_APP_S
```

```
make clean
```

```
make
```

依電腦性能不同，編譯時間 15~30分鐘

得到輸出結果 (*.elf)

```
./obj_epii_evb_icv30_bdv10/gnu_epii_evb_WLCSP65/EPII_CM55M_gnu_epii_evb_WLCSP65_s.elf
```

Executable and
Linkable Format
(**ELF**)

資料來源：

https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2/tree/main/EPII_CM55M_APP_S/app/scenario_app/tflm_yolov8_pose

如何在 Windows 環境編譯 (2/2)

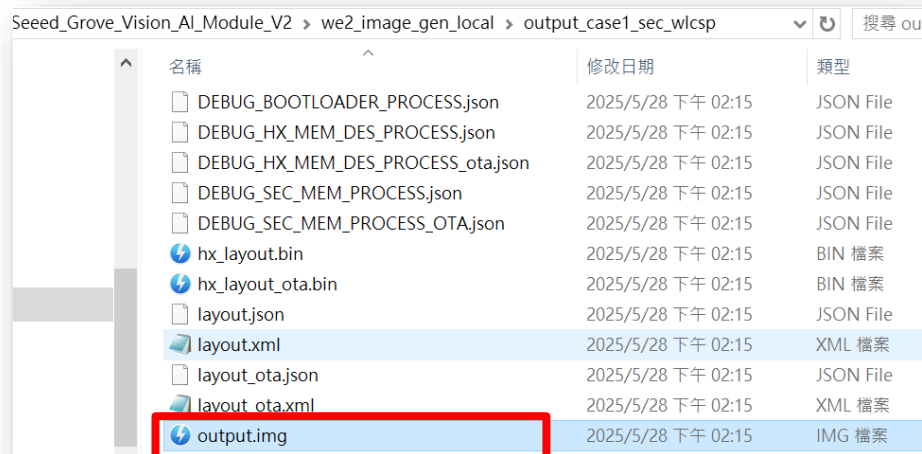
4. 產生燒錄用映像檔 (*.img)

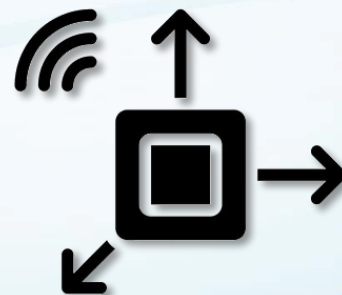
```
cd ../we2_image_gen_local/
```

```
cp ../EPII_CM55M_APP_S/obj_epii_evb_icv30_bdv10/gnu_epii_evb_WLCSP65/EPII_CM55M_gnu_epii_evb_WLCSP65_s.elf input_case1_secboot/
```

```
we2_local_image_gen project_case1_blp_wlcsp.json
```

得到輸出結果 `./output_case1_sec_wlcsp/output.img`





12.4. 模型部署與測試

安裝驅動程式 (USB Virtual COM)

Driver

If you find that the Grove Vision AI V2 is not recognised after connecting it to your computer. Then you may need to install the CH343 driver on your computer. Here are some links to download and install the CH343 driver.

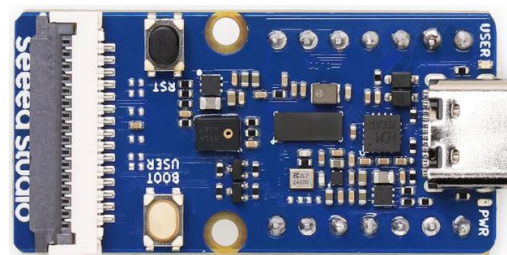
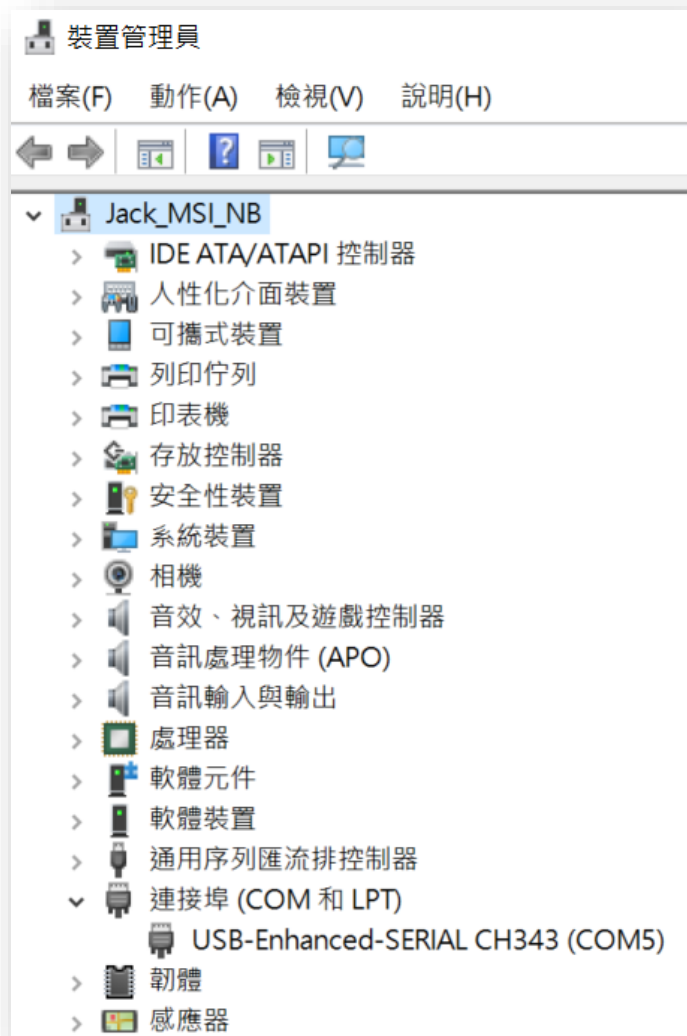
- Windows Vendor VCP Driver One-Click Installer: CH343SER.EXE
- Windows Vendor VCP Driver: CH343SER.ZIP
- Windows CDC driver one-click installer: CH343CDC.EXE
- Windows CDC driver: CH343CDC.ZIP
- macOS Vendor VCP Driver: CH34xSER_MAC.ZIP



只需安裝一次
之後可略過

Windows環境 https://wiki.seeedstudio.com/grove_vision_ai_v2/

開發板組裝 — 連接USB



COM 埠號會變
請依自己電腦上埠號為準

Type C



Type A



Type C



燒錄方式1 — 使用 Edge Impulse CLI

- 安裝Edge Impulse CLI
`npm install -g edge-impulse-cli --force`
- 安裝完成後會得到六項工具
 - `edge-impulse-daemon`, `edge-impulse-uploader`, `edge-impulse-data-forwarder`, `edge-impulse-run-impulse`, `edge-impulse-blocks`
 - **himax-flash-tool** Haimax WE1 & WE2 開發板映像檔燒錄工具
- 切換到*.img所在路徑，執行燒錄
 - `himax-flash-tool -d WiseEye2 -f output.img`
 - 待出現完成訊息後，按下開發板上重置鍵完成燒錄。

<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>

燒錄方式2 — 使用 Python + xmodem

可使用其它類似
COM埠通訊軟體

➤ 下載 Tera Term <https://sourceforge.net/projects/tera-term/>

➤ 啟動 Tera Term ，設定完 COM 埠連接參數後關閉。

➤ 安裝 xmodem

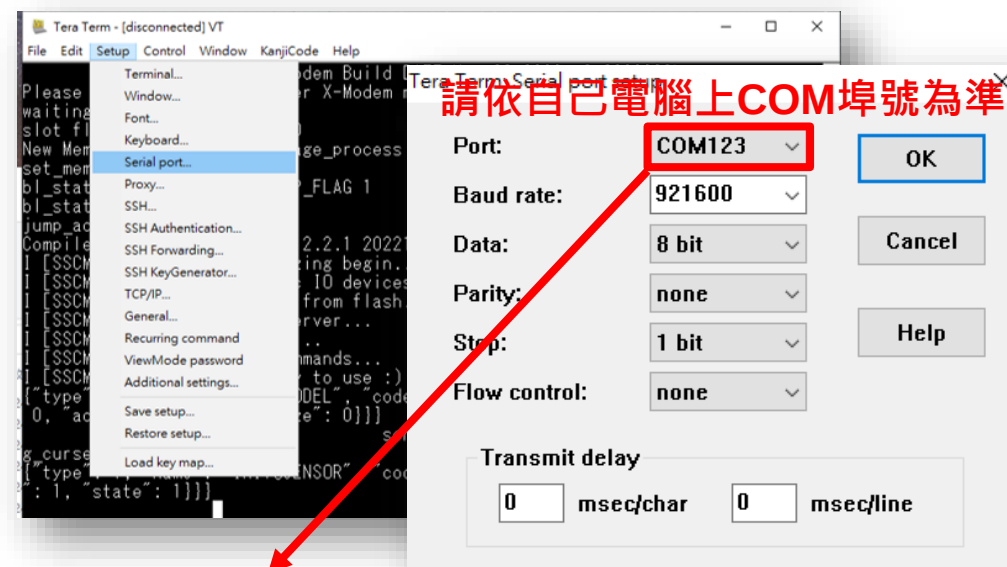
`pip install -r xmodem/requirements.txt`

➤ 插入 Grove Vision AI V2
檢查工作中 COM 埠號

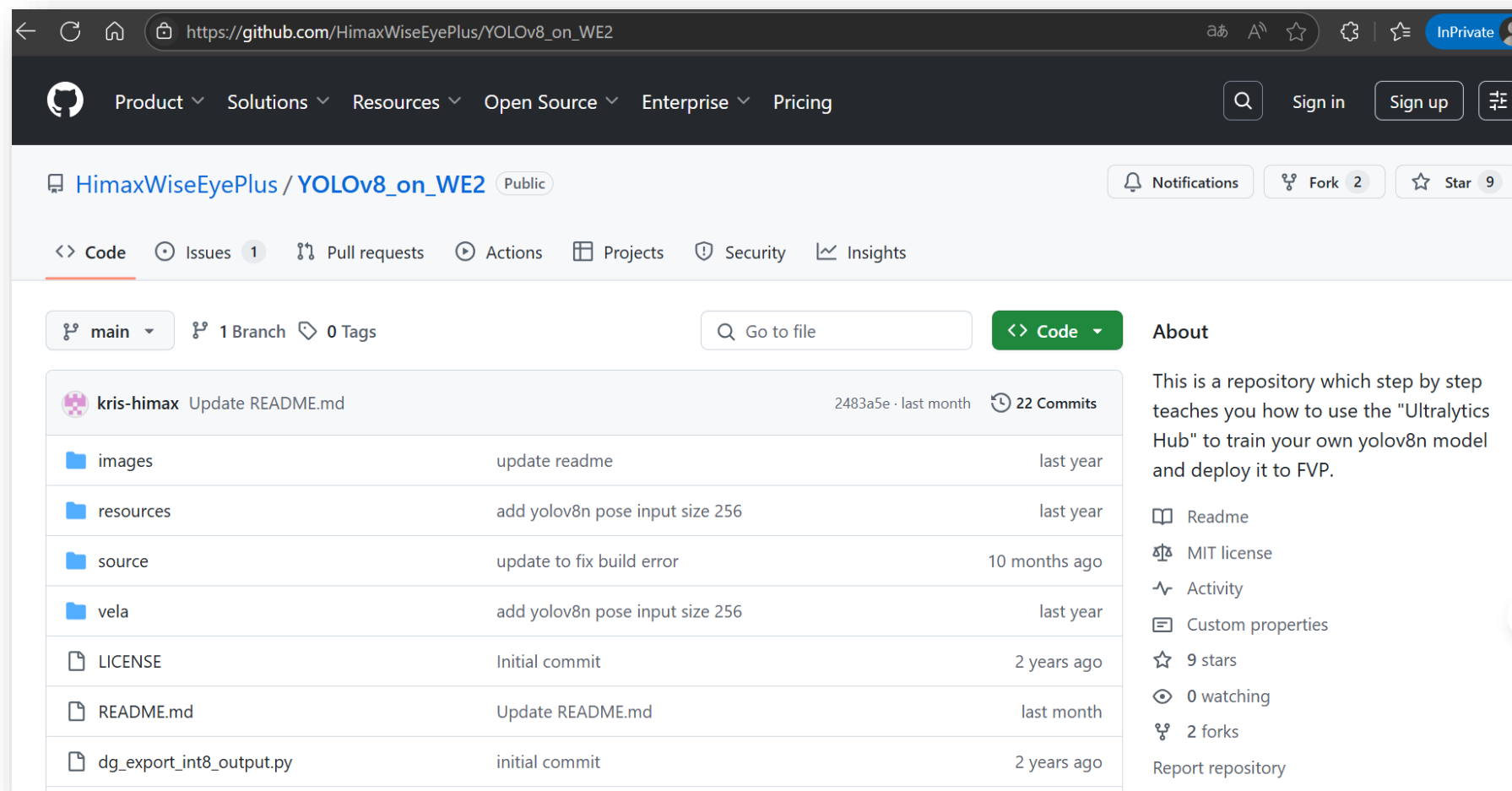
➤ 以 Python 方式燒錄

`python xmodem\xmodem_send.py --port=COM123 --baudrate=921600
--protocol=xmodem --
file=we2_image_gen_local\output_case1_sec_wlcsp\output.img`

➤ 待出現完成訊息後，按下開發板上重置鍵完成燒錄。



Yolov8n on WE2 重新訓練



物件偵測

姿態估測

Vela 轉換

模型部署

https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2

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https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2
- Github – HimaxWiseEyePlus / YOLOv8_on_WE2
https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2

延伸閱讀

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<https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>

- Seeed Studio Grove Vision AI Module V2

https://wiki.seeedstudio.com/grove_vision_ai_v2/

- Seeed SenseCraft Web Toolkit

<https://seeed-studio.github.io/SenseCraft-Web-Toolkit/#/setup/process>

- MJRoBot (Marcelo Rovai), MJRoBot (Marcelo Rovai)Computer Vision at the Edge with Grove Vision AI Module V2

<https://www.hackster.io/mjrobot/computer-vision-at-the-edge-with-grove-vision-ai-module-v2-0003c7>

沒有最邊



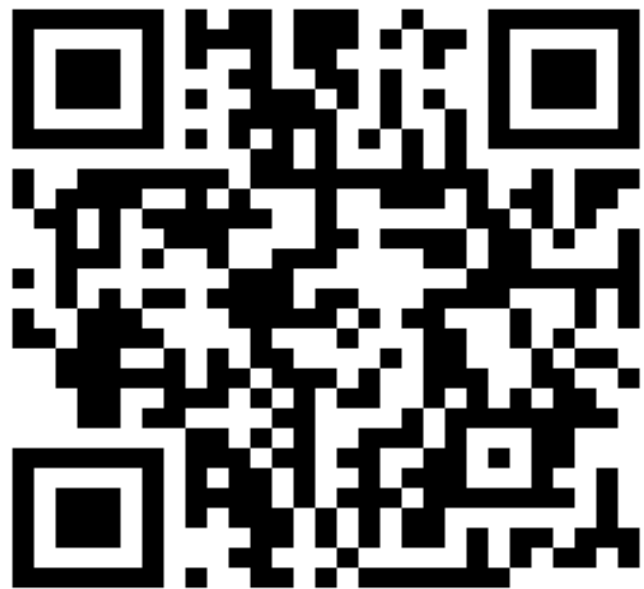
只有更邊

歡迎加入
邊緣人俱樂部



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開 源 : <https://github.com/OmniXRI>

YOUTUBE 直播 : <https://www.youtube.com/@omnixri1784/streams>